



Trial Examination 2024

HSC Year 12 Physics

Solutions and Marking Guidelines

SECTION I**Question 1 A**

A is correct. The speed of light in a vacuum is an absolute constant; it is not dependent on the speed of the light source or the observer.

B, C and D are incorrect. These options do not adhere to Einstein's postulates, which state that the speed of light in a vacuum is constant and all inertial frames of reference are equivalent.

Mod 7 The Nature of Light (PH12-4, 12-7, 12-14)

Bands 2-3

Question 2 B

Using the spectral class classification system OBAFGKM – where O represents stars with the highest surface temperatures and M represents stars with the lowest surface temperatures – star Y (B5) has the highest surface temperature, followed by star Z (A2), and then star X (M2). The data relating to the stars' distances from the Sun and apparent magnitudes is not relevant to surface temperature.

Mod 8 From the Universe to the Atom (PH12-5, 12-6, 12-15)

Bands 2-3

Question 3 C

C is correct. When the paintball is shot horizontally, it has an initial horizontal component of velocity that is equal to its launch velocity. The horizontal component remains unchanged throughout the paintball's motion ($V_x = \text{constant} = 80 \text{ m s}^{-1}$).

A is incorrect. This option may be reached by calculating $\frac{80}{10} = 8.0$.

B is incorrect. This option may be reached by calculating $\frac{80}{9.8} = 8.2$.

D is incorrect. This option may be reached by calculating $10 \times 80 = 800$.

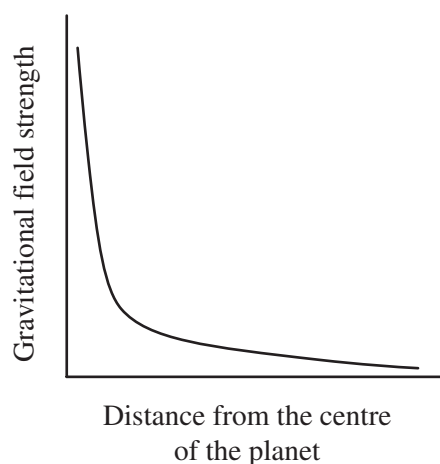
Mod 5 Advanced Mechanics (PH12-4, 12-5, 12-6, 12-12)

Bands 2-3

Question 4 D

D is correct. The graph of gravitational field strength versus mass of the planet would show that gravitational field strength increases at a steady rate as mass increases.

A is incorrect. The graph of gravitational field strength versus distance from the centre of the planet would show an inverse square relationship. There would be an initial steep decline in gravitational field strength that slows down, proportional to the distance from the centre of the planet, as shown in the graph.



B and **C** are incorrect. The title of the x -axis could be inverse square of the distance from the centre of the planet because gravitational field strength increases proportionally to the inverse square of the distance from the centre of the planet. Furthermore, Option **B** refers to the surface, rather than the centre, of the planet; and option **C** refers to the square, rather than the inverse square.

Mod 5 Advanced Mechanics (PH12–4, 12–5, 12–7, 12–12)

Bands 2–3

Question 5 A

A is correct. There are 1 000 000 μs in 1 s. Reading from the graph, 50% of the sample remains at approximately 2.0 μs . Therefore, the half-life of the sample in seconds is $\frac{2.0}{1\,000\,000} = 2.0 \times 10^{-6}$ s.

B is incorrect. This option may be reached by misreading the graph and not converting the half-life to seconds.

C is incorrect. This option may be reached by calculating $\frac{2.0}{1000} = 2.0 \times 10^{-3}$ s.

D is incorrect. This option is a misinterpretation of the graph that assumes that the axis for time is already given in seconds.

Mod 8 From the Universe to the Atom (PH12–5, 12–15)
Working Scientifically Skills (PH12–6)

Bands 2–3

Question 6 D

D is correct. According to Einstein's mass–energy equivalence relationship, $E = mc^2$, mass is a form of energy.

A and **C** are incorrect. Mass and energy are neither independent nor identical, as they are different forms of the same underlying quantity.

B is incorrect. An object's rest mass is the same for all inertial frames of reference.

Mod 8 From the Universe to the Atom (PH12–5, 12–6, 12–14)

Bands 3–4

Question 7 D

$$F = \frac{GMm}{r^2}$$

$$F \propto \frac{M}{r^2}$$

Given that the distance between the objects was decreased by one third, $r = 1 - \frac{1}{3} = \frac{2}{3}$.

Given that the mass of one of the objects was increased by a factor of 4:

$$\begin{aligned} F &= \frac{4}{\left(\frac{2}{3}\right)^2} \\ &= \frac{4}{\frac{4}{9}} \\ &= 9 \end{aligned}$$

Thus, $F_1 = 9F$.

Mod 5 Advanced Mechanics (PH12–4, 12–5, 12–6, 12–12)

Bands 4–5

Question 8 **C**

C is correct and **D** is incorrect.

$$F = qvB$$

$$= 1.602 \times 10^{-19} \times 1200 \times 3.00$$

$$= 5.7672 \times 10^{-16} \text{ N to the right}$$

It can be determined that the direction is to the right as the proton enters the field because the particle has a positive charge and the magnetic field is out of the page. This is according to the right-hand grip rule.

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\vec{a} = \frac{\vec{F}}{m}$$

$$= \frac{5.7672 \times 10^{-16}}{1.673 \times 10^{-27}}$$

$$= 3.45 \times 10^{11} \text{ m s}^{-2} \text{ to the right}$$

A and **B** are incorrect. These options do not calculate $\vec{F} = m\vec{a}$ or account for the mass of the proton.

Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)

Bands 3–4

Question 9 **D**

D is correct.

$$E = mc^2$$

$$= 3.07 \times 10^{-28} \times (3.00 \times 10^8)^2$$

$$= 2.763 \times 10^{-11} \text{ J}$$

$$\frac{2.763 \times 10^{-11}}{1.602 \times 10^{-19}} = 1.72 \times 10^8 \text{ eV}$$

A is incorrect. This option calculates $(2.763 \times 10^{-11}) \times (1.602 \times 10^{-19}) = 4.42 \times 10^{-30}$.

B is incorrect. This option calculates $(3.99 \times 10^{-25}) - (3.07 \times 10^{-28}) = 3.99 \times 10^{-25}$.

C is incorrect. This option calculates the energy in joules but uses the incorrect unit of measurement.

Mod 8 From the Universe to the Atom (PH12–5, 12–6, 12–15)

Bands 3–4

Question 10 **A**

$$m = 2200 \text{ kg}$$

$$v = 10.5 \text{ m s}^{-1}$$

$$r = 23 \text{ m}$$

$$F_c = \frac{mv^2}{r}$$

$$= \frac{2200 \times 10.5^2}{23}$$

$$= 10\,546 \text{ N}$$

The magnitude of the frictional force in all conditions is less than 10 546 N; therefore, it is not safe to drive around the bend under any of the conditions.

Mod 5 Advanced Mechanics (PH12–5, 12–6, 12–12)

Bands 4–5

Question 11 **A**

When bismuth-211 undergoes one alpha decay, its mass number decreases by 4, and its atomic number decreases by 2, which indicates the release of a helium nucleus. Therefore, the daughter nuclide produced has a mass number of $211 - 4 = 207$ and an atomic number of $83 - 2 = 81$. Reading from the periodic table, it can be determined that the daughter nuclide is $^{207}_{81}\text{Tl}$.

Mod 8 From the Universe to the Atom (PH12–6, 12–15)

Bands 2–3

Question 12 **A**

The proton–proton chain is a three-step process in which protons join to form a helium nucleus. It is a relatively slow process and occurs in cooler main sequence stars.

The carbon–nitrogen–oxygen (CNO) cycle is a six-step process in which protons fuse into a helium nucleus using carbon as a catalyst. It is a relatively fast process and occurs in more massive and, therefore, hotter main sequence stars.

Mod 8 From the Universe to the Atom (PH12–5, 12–15)

Bands 3–4

Question 13 **C**

Using the right-hand palm rule:

- The magnetic force acting on LM is perpendicular into the page because the current is moving down the page and the magnetic field (B) is to the left.
- The magnetic force acting on MN is perpendicular into the page because the current is moving down the page and B is to the left.
- There is no magnetic force acting on NO as it is parallel to B .

Mod 6 Electromagnetism (PH12–5, 12–13)

Bands 3–4

Question 14 B

The current in coil *J* flows clockwise when viewed from the left. According to Lenz's law, the induced current in coil *K* that flows through resistor *xy* opposes the change in flux through the coil.

After change I, in which switch *S* is opened and the current stops flowing, a current is induced in coil *K* that flows from *x* to *y*.

After change II, in which coil *K* is brought closer to coil *J* and switch *S* is closed, the current begins to flow through coil *J* again. The magnetic field around coil *J* interacts with the magnetic field induced around coil *K*. This induces a current in the direction that opposes the change. Therefore, the induced current flows from *y* to *x*.

After change III, in which the resistance of *R* is decreased while switch *S* remains closed, the magnetic fields interact in the same way as after change II. Therefore, the direction of the current does not change and flows from *y* to *x*.

Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)

Bands 4–5

Question 15 A

A is correct and **B** is incorrect.

Force acting on the object due to Earth:

$$\begin{aligned}
 mg &= \frac{GMm}{r^2} \\
 g &= \frac{GM}{r^2} \\
 &= \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{\left(\frac{1}{2} \times 3.84 \times 10^8\right)^2} \\
 &= 1.08 \dots \times 10^{-2} \text{ m s}^{-2}
 \end{aligned}$$

Force acting on the object due to the Moon:

$$\begin{aligned}
 mg &= \frac{GMm}{r^2} \\
 g &= \frac{GM}{r^2} \\
 &= \frac{6.67 \times 10^{-11} \times 7.35 \times 10^{22}}{\left(\frac{1}{2} \times 3.84 \times 10^8\right)^2} \\
 &= 1.32 \dots \times 10^{-4} \text{ m s}^{-2}
 \end{aligned}$$

The object's acceleration due to gravity:

$$\begin{aligned}
 g &= (1.08 \dots \times 10^{-2}) - (1.32 \dots \times 10^{-4}) \\
 &= 1.07 \times 10^{-2} \text{ m s}^{-2} \text{ towards Earth}
 \end{aligned}$$

C and **D** are incorrect. These options calculate the force acting on the object due to Earth.

Mod 5 Advanced Mechanics (PH12–4, 12–5, 12–6, 12–12)

Bands 4–5

Question 16 A

A is correct and **B** is incorrect.

Magnitude of the force acting on wire *X* due to current I_Y (2.0 A):

$$\begin{aligned}\frac{F}{l} &= \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r} \\ &= \frac{4\pi \times 10^{-7}}{2\pi} \frac{1.0 \times 2.0}{5 \times 10^{-3}} \\ &= 8 \times 10^{-5} \text{ N m}^{-1} \text{ up}\end{aligned}$$

Magnitude of the force acting on wire *X* due to current I_Z (3.0 A):

$$\begin{aligned}\frac{F}{l} &= \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r} \\ &= \frac{4\pi \times 10^{-7}}{2\pi} \frac{1.0 \times 3.0}{10 \times 10^{-3}} \\ &= 6 \times 10^{-5} \text{ N m}^{-1} \text{ down}\end{aligned}$$

Magnitude of the net force acting on wire *X*:

$$\begin{aligned}F_{\text{net}} &= (8 \times 10^{-5}) - (6 \times 10^{-5}) \\ &= 2 \times 10^{-5} \text{ N m}^{-1} \text{ up}\end{aligned}$$

The directions are determined by the right-hand grip rule.

C is incorrect. This option calculates the magnitude of the force acting on wire *X* due to the current in wire *Y* only.

D is incorrect. This option calculates the magnitude of the force acting on wire *X* due to the current in wire *Y* only and does not take into account that the current in wire *Y* is in the opposite direction to the current in wire *X*.

Mod 6 Electromagnetism (PH12–5, 12–13)
Working Scientifically Skills (PH12–6)

Bands 4–5

Question 17 C

$$\begin{aligned}l &= l_0 \sqrt{1 - \frac{v^2}{c^2}} \\ 15.0 &= l_0 \sqrt{1 - \frac{(0.450c)^2}{c^2}} \\ l_0 &= \frac{15.0}{\sqrt{1 - 0.2025}} \\ &= 16.8 \text{ m}\end{aligned}$$

The pilot would measure the length of the UFO to be 16.8 m. They would measure the width of the UFO to be 3.50 m, the same measurement as the observer on Earth, because only the dimension that is parallel to the velocity, in this case the length of the UFO, is contracted.

Mod 7 The Nature of Light (PH12–4, 12–14)

Bands 5–6

Question 18 B

B is correct and **A** is incorrect.

$$\begin{aligned}
 F &= I l B \sin \theta \\
 &= 0.050 \times 2 \times 1.0 \times \sin(60^\circ) \\
 &= 0.087 \text{ N}
 \end{aligned}$$

According to the right-hand palm rule, since the magnetic field is left to right, the force is out of the page.

C and **D** are incorrect. These options do not convert the length to metres before calculating the force.

Mod 6 Electromagnetism (PH12-4, 12-5, 12-13)

Bands 4-5

Question 19 A

A is correct and **B** is incorrect. The speed of the satellite would need to increase in the lower orbit.

$$\begin{aligned}
 v_{\text{at 850 km orbit}} &= \sqrt{\frac{GM}{r}} \\
 &= \sqrt{\frac{(6.67 \times 10^{-11}) \times (6.0 \times 10^{24})}{(6.371 \times 10^6) + (0.85 \times 10^6)}} \\
 &= \sqrt{\frac{4.002 \times 10^{14}}{7.221 \times 10^6}} \\
 &= 7444.57 \dots \text{ m s}^{-1} \\
 v_{\text{at 650 km orbit}} &= \sqrt{\frac{GM}{r}} \\
 &= \sqrt{\frac{(6.67 \times 10^{-11}) \times (6.0 \times 10^{24})}{(6.371 \times 10^6) + (0.65 \times 10^6)}} \\
 &= \sqrt{\frac{4.002 \times 10^{14}}{7.021 \times 10^6}} \\
 &= 7549.86 \dots \text{ m s}^{-1}
 \end{aligned}$$

Change in speed:

$$\begin{aligned}
 \Delta v &= 7549.86 \dots - 7444.57 \dots \\
 &= 105 \text{ m s}^{-1}
 \end{aligned}$$

C and **D** are incorrect. These options do not include Earth's radius in the calculations.

Mod 5 Advanced Mechanics (PH12-4, 12-5, 12-6, 12-12)

Bands 5-6

Question 20 **C**

C is correct and **D** is incorrect. Both Exo and Endo experience acceleration due to gravity according

to $g = \frac{GM}{r^2}$. Exo is approximately 2.5 times further away from the star than Endo is $\left(\frac{1.56 \times 10^9}{6.2 \times 10^8} = 2.5 \right)$

and so will experience a lower acceleration due to gravity than Endo. In accordance with the inverse square

relationship $\left(g \propto \frac{1}{r^2} \right)$, Endo will experience an acceleration due to gravity that is approximately $\frac{1}{2.5^2}$,

or 6.3, times greater than the acceleration due to gravity experienced by Exo.

A and **B** are incorrect. These options do not use the inverse square relationship to calculate the acceleration due to gravity.

Mod 5 Advanced Mechanics (PH12–4, 12–5, 12–6, 12–12)

Bands 5–6

SECTION II

Question 21

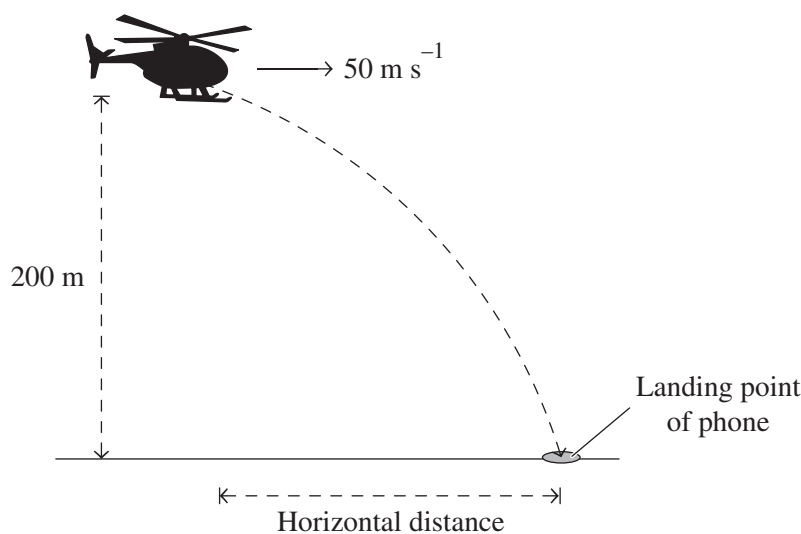
- (a) $u = 0$
 $a = 9.8 \text{ m s}^{-2}$
 $t = 3.5 \text{ s}$
 $s = ?$
 $s = ut + \frac{1}{2}at^2$
 $= 0 + \frac{1}{2} \times 9.8 \times 3.5^2$
 $= 60 \text{ m}$

Mod 5 Advanced Mechanics (PH12–6, 12–12)

Bands 2–3

- Provides the correct solution using the required variables AND the correct formula. 2
- Provides some relevant working 1

- (b) The scenario can be illustrated as follows.



$$a_y = 9.8 \text{ m s}^{-2}$$

$$a_x = 0 \text{ m s}^{-1}$$

$$u = 50 \text{ m s}^{-1}$$

The angle of projection is 0° because the phone was dropped from the helicopter. Therefore:

$$u_x = u \cos \theta$$

$$= 50 \cos(0)$$

$$= 50 \text{ m s}^{-1}$$

$$u_y = u \sin \theta$$

$$= 50 \sin(0)$$

$$= 0 \text{ m s}^{-1}$$

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Calculating the phone's time of flight gives:

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$-200 = 0 + \frac{1}{2}(-9.8) \times t^2$$

$$t^2 = \frac{-200}{-4.9}$$

$$t = \sqrt{40.81 \dots}$$

$$= 6.38 \dots \text{ seconds}$$

Therefore, calculating the horizontal distance that the phone travels gives:

$$s_x = u_x t$$

$$= 50.0 \times 6.38 \dots$$

$$= 319 \text{ m}$$

Note: Responses are not required to include diagrams to obtain full marks.

Mod 5 Advanced Mechanics (PH12–6, 12–12)

Bands 3–4

- Provides the correct solution. 3

- Calculates the phone's time of flight. 2

- Provides some relevant working 1

Question 22(a) negatively charged (*as B is into the page and the particle arcs to the right*)

Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)

Bands 2–3

- States that the particle is negatively charged. 1

(b) $\frac{mv^2}{r} = qvB$

$$r = \frac{mv}{qB}$$

$$\frac{q}{m} = \frac{v}{rB}$$

$$= \frac{2.50 \times 10^4}{0.050 \times 1.95 \times 10^{-6}}$$

$$= 2.6 \times 10^{11} \text{ C kg}^{-1}$$

Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)

Bands 4–5

- Substitutes into the relevant formula correctly 3

- Attempts to substitute into the relevant formula 2

- Identifies the relevant formula. 1

Question 23

- (a) When the polarised sunglasses are placed in front of the polarised light source, the sunglasses will cause the light to dim. The student could place one of the pairs of sunglasses in front of one of the light sources, then rotate the pair of sunglasses through different angles until they have been rotated 90° . The student should repeat this process for each pair of sunglasses with each light source.

Mod 7 The Nature of Light (PH12–2, 12–7, 12–14)

Bands 3–4

- Outlines a simple procedure.
AND
- Explains the expected results. 3

- Outlines a simple procedure 2

- Provides some relevant information 1

- (b) If light passes through one polariser and then passes through a second polariser that is perpendicular to the first, then none of the light that passes through the first polariser will pass through the second. This cannot be explained by Newton's particle model of light because it describes light as a stream of particles, which should pass through both polarisers. It is also not possible to explain polarisation with Huygens' longitudinal wave model. Longitudinal waves cannot be polarised because the direction of oscillation of a longitudinal wave is parallel, not perpendicular, to the direction of propagation.

Polarisation supports the transverse wave model of light. According to this model, light waves are electromagnetic waves, which have an electric field and a magnetic field that oscillate perpendicular to each other. This means that the electric and magnetic fields can pass through or be blocked by a polariser depending on its polarisation axis.

Mod 7 The Nature of Light (PH12–7, 12–14)

Bands 5–6

- Outlines polarisation.
AND
- Explains that polarisation cannot be explained by the particle model of light.
AND
- Explains that polarisation cannot be explained by the longitudinal wave model of light.
AND
- Outlines electromagnetic waves.
AND
- Explains how polarisation supports the transverse wave model of light. 5

- Any FOUR of the above points. 4

- Any THREE of the above points. 3

- Any TWO of the above points. 2

- Provides some relevant information 1

Question 24

(a) $d \sin \theta = m \lambda$
 $3.0 \times 10^{-5} \times \sin \theta = 1 \times (500 \times 10^{-9})$
 $\sin \theta = \frac{500 \times 10^{-9}}{3.0 \times 10^{-5}}$
 $\theta = 0.95 \dots^\circ$
 $\tan(0.95 \dots^\circ) = \frac{x}{6.0}$
 $x = 0.10 \text{ m}$

Mod 7 The Nature of Light (PH12–2, 12–4)

Bands 5–6

- Calculates the distance between each of the maxima2

- Calculates the angle.1

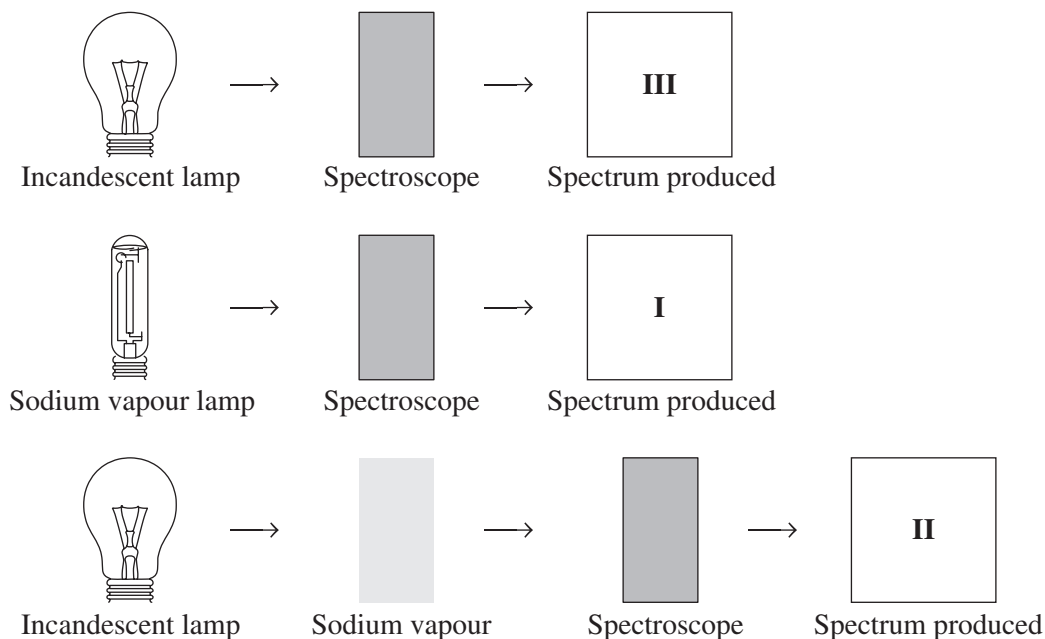
- (b) The experiment supports the wave model of light because the light wave diffracts as it passes through the slits. This is seen in the diffraction pattern on the screen; the maxima are due to constructive interference of the diffracted light, and the minima are due to destructive interference.

Mod 7 The Nature of Light (PH12–2, 12–4, 12–7, 12–14)

Bands 5–6

- States that the experiment supports the wave model of light.
AND
- Explains why the experiment supports the wave model of light with reference to diffraction.2

- States that the experiment supports the wave model of light1

Question 25

The first incandescent lamp will produce spectrum III, which is a continuous spectrum, as its heated filament produces all wavelengths of light. As light is produced by an electric current passing through a conductor with a high resistance and melting point, the collision between the electrons and the conductor produces heat and light.

The sodium vapour lamp will produce spectrum I, which is an emission spectrum. The two yellow lines on the black background correspond to photons of light that are released when the energy levels of the excited electrons in the sodium vapour decrease.

The second incandescent lamp will produce spectrum II, which is an absorption spectrum. The lamp produces all wavelengths of light. Some of these wavelengths will be absorbed by the sodium vapour and others will pass through. The wavelengths that are absorbed will appear as black lines on a coloured background.

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Mod 7 The Nature of Light (PH12–4, 12–7, 12–14)

Bands 5–6

- Identifies which spectrum (I, II or III) will be produced by each of the THREE light sources.
AND
- Identifies which type of spectrum (continuous, absorption or emission) will be produced by each of the THREE light sources.
AND
- Provides a detailed explanation for ALL THREE identified spectrum types 7

- Identifies which spectrum (I, II or III) will be produced by each of the THREE light sources.
AND
- Identifies which type of spectrum (continuous, absorption or emission) will be produced by each of the THREE light sources.
AND
- Provides a detailed explanation for TWO of the identified spectrum types 6

- Identifies which spectrum (I, II or III) will be produced by each of the THREE light sources.
AND
- Identifies which type of spectrum (continuous, absorption or emission) will be produced by any TWO light sources.
AND
- Provides a detailed explanation for TWO of the identified spectrum types 5

- Identifies which spectrum (I, II or III) will be produced by each of the THREE light sources.
AND
- Identifies which type of spectrum (continuous, absorption or emission) will be produced by any TWO light sources.
AND
- Provides a detailed explanation for ONE of the identified spectrum types. 4

- Identifies which spectrum (I, II or III) will be produced by any TWO light sources.
AND
- Identifies which type of spectrum (continuous, absorption or emission) will be produced by any TWO light sources. 3

- Identifies which spectrum (I, II or III) will be produced by any TWO light sources.
AND
- Identifies which type of spectrum (continuous, absorption or emission) will be produced by any ONE light source. 2

- Provides some relevant information 1

Question 26

Hubble's law states that the further a galaxy is from Earth's galaxy, the faster it is moving away from Earth. Hubble plotted velocity against distance and obtained a linear graph, indicating that the universe is expanding. This can be seen in the red shift of the spectra. Comparing the spectrum of the Sun to the spectrum of light from a distant galaxy shows that the Sun and light from the distant galaxy have the same spectral lines; however, in the spectrum of light from the distant galaxy, these lines have shifted towards the red end. This indicates a longer wavelength and thus a greater distance from Earth, thereby providing evidence that the universe is expanding. This subsequently supports the Big Bang theory, which proposes that all matter and energy in the universe was originally contained to a single extremely hot and dense point before rapidly expanding.

Mod 8 From the Universe to the Atom (PH12-5, 12-6, 12-7, 12-15)

Bands 4-5

- States Hubble's law.
AND
- Identifies the red shift in the spectra.
AND
- Explains how the spectra support Hubble's law.
AND
- Explains how the spectra support the Big Bang theory. 4

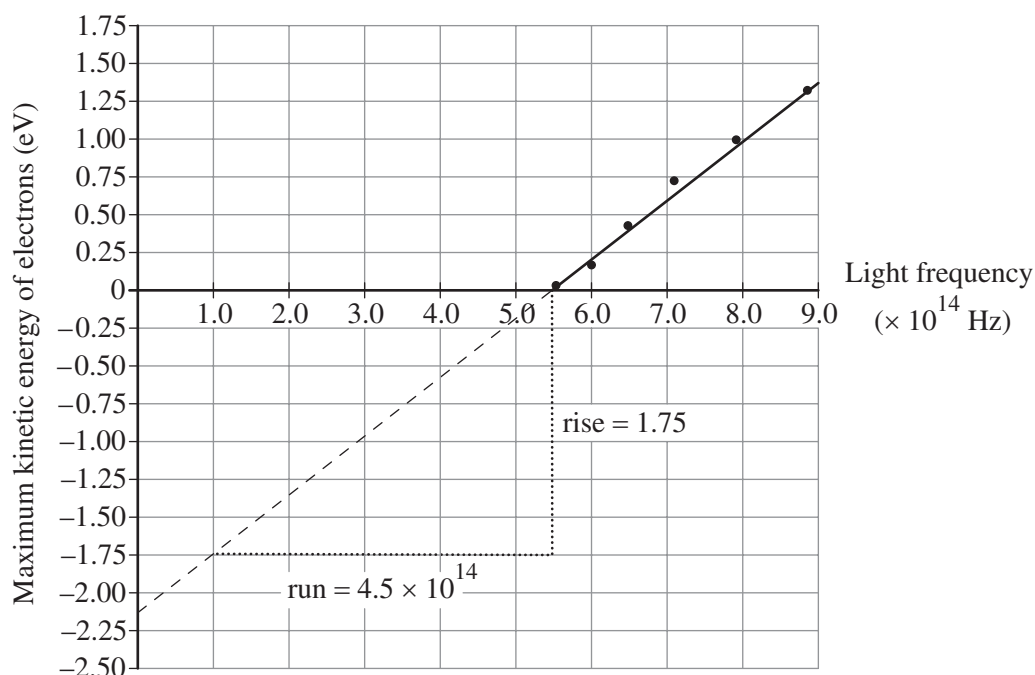
- Any THREE of the above points. 3

- Any TWO of the above points. 2

- Provides some relevant information 1

Question 27

(a)



$\phi = 2.13$ eV (based on extrapolation of the line of best fit and identifying the y-intercept)

$$\begin{aligned}
 h &= \frac{\text{rise}}{\text{run}} \\
 &= \frac{0 - (-1.75)}{(5.5 \times 10^{14} - 1.0 \times 10^{14})} \\
 &= \frac{1.75}{4.5 \times 10^{14}} \\
 &= 3.9 \times 10^{-15} \text{ eV s}
 \end{aligned}$$

Note: For the work function, ϕ , accept responses in the range 2.10–2.30 eV. For Planck's constant, h , accept responses in the range $3.5\text{--}4.5 \times 10^{-15}$ eV s.

Mod 7 The Nature of Light (PH12–4, 12–7, 12–14)

Bands 2–3

- Plots all data points AND draws a line of best fit.
AND
- Determines ϕ of the metal by extrapolating the line.
AND
- Calculates h by calculating the gradient of the line. 3

- Any TWO of the above points. 2

- Any ONE of the above points 1

$$\begin{aligned}
 \text{(b)} \quad K_{\max} &= \frac{hc}{\lambda} - \phi \\
 &= \frac{3.9 \times 10^{-15} \times 3.00 \times 10^8}{4.20 \times 10^{-7}} - 2.13 \\
 &= 0.66 \text{ eV}
 \end{aligned}$$

Thus, as there is energy remaining from the photon after the electron has escaped the metal, the photoelectric effect will occur.

*Note: Consequential on answer to **Question 27(a)**.*

Mod 7 The Nature of Light (PH12–4, 12–7, 12–14)

Bands 4–5

- Calculates the maximum kinetic energy of the electrons emitted.
AND
- Determines that the photoelectric effect will occur. 2

- Calculates the maximum kinetic energy of the electrons emitted. 1

Question 28

For conclusion I to be true, light from Einstein's face would be travelling at the speed of light (c) relative to the train. A stationary observer outside the train would see the light travelling at twice the speed of light, $2c$. Seeing his reflection does not violate the principle of relativity, as the speed of light is constant in all inertial frames of reference. Therefore, conclusion I is correct.

For conclusion II to be true, light from Einstein's face would not be travelling at the speed of light (c) and so would never reach the mirror. Therefore, he would not see his reflection. This conclusion violates the principle of relativity and so is incorrect.

Mod 7 The Nature of Light (PH12–4, 12–7, 12–14)

Bands 5–6

- Discusses conclusion I AND conclusion II in detail.
AND
- States that conclusion I is correct 4

- Discusses conclusion I AND conclusion II.
AND
- States that conclusion I is correct 3

- Discusses conclusion I OR conclusion II.
AND
- States conclusion I is correct 2

- Discusses conclusion I OR conclusion II with limited detail.
OR
- States that conclusion I is correct 1

Question 29

The fundamental particles of the Standard Model of matter are quarks, leptons, bosons and the Higgs boson. Experiments using particle accelerators have provided evidence for the existence of these particles and three of the fundamental forces; weak nuclear force, strong nuclear force and electromagnetism. Particle accelerators allow particles to collide at great speeds that produce large amounts of energy. This causes the particles to break down which makes it possible to determine their composition.

In the Bohr model of the atom, the fundamental particles are protons, neutrons and electrons. However, particle accelerators have led to the discovery that protons and neutrons are not fundamental particles, as they can be broken down into quarks. The Standard Model identifies six types of quarks: up, down, charm, strange, top and bottom. Experiments using particle accelerators to observe electron–proton collisions have led to the discovery that protons comprise two up quarks and one down quark; and neutrons comprise one up quark and two down quarks.

The Standard Model also identifies six leptons: electron, muon, tau, electron neutrino, muon neutrino and tau neutrino. An experiment using a particle accelerator that enabled electron–positron collisions led to the discovery of the tau lepton, which subsequently implied the existence of the tau neutrino lepton. Further research involving particle acceleration has also provided evidence of neutrinos.

Furthermore, particle accelerators such as the Large Hadron Collider (LHC) have been key to confirming the existence of bosons (including photons, gluons, W bosons, Z bosons and the Higgs boson). In the Standard Model, bosons are force carriers, which hold together matter particles. The photon carries the electromagnetic force, the gluon carries the strong nuclear force, and the W and Z bosons carry the weak nuclear force. The use of particle accelerators has increased understanding of these fundamental forces.

Over time, the technology of particle accelerators has improved, enabling the ongoing identification and analysis of many components of the Standard Model.

Mod 8 From the Universe to the Atom (PH12–5, 12–6, 12–7, 12–15)

Bands 5–6

- Explains the Standard Model of matter.
AND
 - Explains how particle accelerators work.
AND
 - Analyses how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
AND
 - Analyses how particle accelerators have provided evidence of the fundamental forces of the Standard Model8
-
- Outlines the Standard Model of matter.
AND
 - Outlines how particle accelerators work.
AND
 - Analyses how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
AND
 - Analyses how particle accelerators have provided evidence of the fundamental forces of the Standard Model7

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(continued)

- Outlines the Standard Model of matter.
AND
 - Outlines how particle accelerators work.
AND
 - Outlines how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
AND
 - Outlines how particle accelerators have provided evidence of the fundamental forces of the Standard Model6
-
- Outlines the Standard Model of matter OR how particle accelerators work.
AND
 - Outlines how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
AND
 - Outlines how particle accelerators have provided evidence of the fundamental forces of the Standard Model.5
-
- Outlines how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
AND
 - Outlines how particle accelerators have provided evidence of the fundamental forces of the Standard Model4
-
- Identifies how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
AND
 - Identifies how particle accelerators have provided evidence of the fundamental forces of the Standard Model.3
-
- Identifies how particle accelerators have provided evidence of the fundamental particles of the Standard Model.
OR
 - Identifies how particle accelerators have provided evidence of the fundamental forces of the Standard Model2
-
- Provides some relevant information1

Question 30

(a) $V_p I_p = V_s I_s$

$$V_s = \frac{V_p I_p}{I_s}$$

$$= \frac{120 \times 3.0}{30}$$

$$= 12 \text{ V}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$N_p = \frac{V_p N_s}{V_s}$$

$$= \frac{120 \times 5}{12}$$

$$= 50 \text{ turns}$$

Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)
Working Scientifically Skills (PH12–6)

Bands 3–4

- Calculates V_s .
AND
- Determines N_p 3

- Calculates V_s .
OR
- Determines N_p using ratios2

- Provides some relevant working1

(b) *For example:*

Faraday's law states that a change in magnetic flux will induce an electromotive force (emf), which will in turn induce a current.

When the switch is left open, there is no magnetic field; when it is left closed, there is a constant magnetic field. Therefore, coil *B* does not experience a change in flux, so the compass needle does not deflect.

When the switch is initially opened or closed, the magnetic field produced by coil *A* rapidly changes, causing coil *B* to experience a change in magnetic flux. This induces a current in coil *B*, which in turn causes the compass needle to deflect.

Mod 6 Electromagnetism (PH12–4, 12–5, 12–7, 12–13)

Working Scientifically Skills (PH12–6)

Bands 5–6

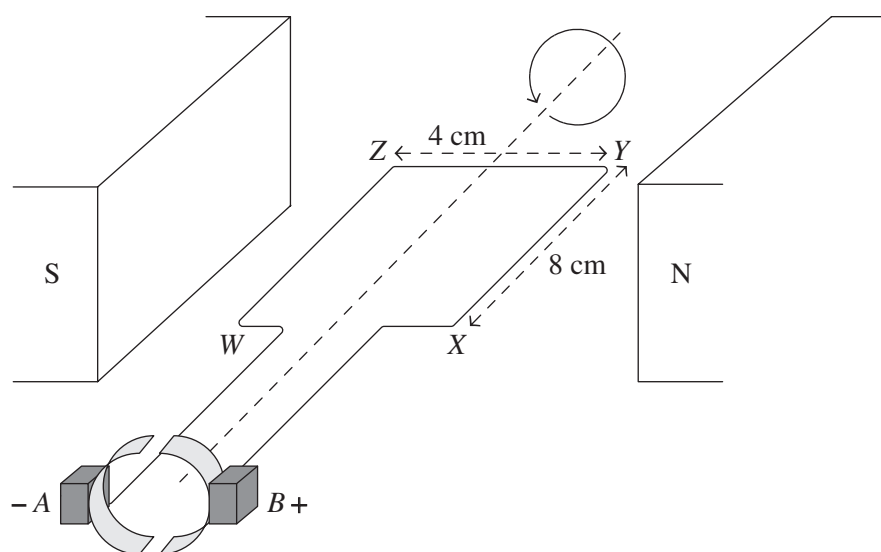
- Explains what happens in coil *B* when the switch is initially opened or closed.
AND
- Explains why the needle deflects when the switch is initially opened or closed.
AND
- Explains what happens in coil *B* when the switch is left open or closed 3

- Outlines what happens in coil *B* when the switch is initially opened or closed.
AND
- Outlines what happens in coil *B* when the switch is left open or closed. 2

- Provides some relevant information 1

Question 31

(a)



Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)

Bands 2–3

- Labels terminal *A* as negative and terminal *B* as positive 1

- (b) The commutator (split-ring) reverses the direction of the current every half rotation. If the current is not reversed every half rotation, the forces on *WZ* and *XY* will remain in a constant direction, resulting in a torque that reverses.

Mod 6 Electromagnetism (PH12–4, 12–5, 12–13)

Bands 3–4

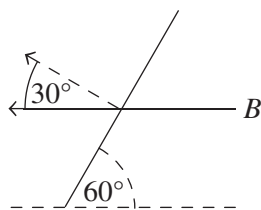
- Identifies the commutator.
AND
 - Explains how the commutator maintains the constant torque.
AND
 - Refers to the diagram. 3
-
- Identifies the commutator.
AND
 - Explains how the commutator maintains the constant torque 2
-
- Identifies the commutator.
OR
 - Provides some relevant information 1

(c) Position shown in the diagram (position I):

$$\begin{aligned}\tau &= nIAB \sin \theta \\ &= 1 \times 12 \times 0.080 \times 0.040 \times 0.65 \times \sin(90^\circ) \\ &= 0.025 \text{ N m anticlockwise}\end{aligned}$$

Position where the plane of the loop makes an angle of 60° from the horizontal (position II):

View from front

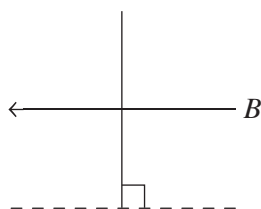


$$\therefore \sin(30^\circ)$$

$$\begin{aligned}\tau &= 1 \times 12 \times 0.080 \times 0.040 \times 0.65 \times \sin(30^\circ) \\ &= 0.012 \text{ N m anticlockwise}\end{aligned}$$

Position where the plane of the loop makes an angle of 90° from the horizontal (position III):

View from front



$$\therefore \sin(0^\circ)$$

$$\begin{aligned}\tau &= 1 \times 12 \times 0.080 \times 0.040 \times \sin(0^\circ) \\ &= 0 \text{ N m anticlockwise}\end{aligned}$$

Note: Responses are not required to include diagrams to obtain full marks.

Mod 6 Electromagnetism (PH 12–4, 12–5, 12–13)

Bands 4–5

- Determines the magnitude of the torque in all THREE positions.
AND
- Determines the direction of rotation in all THREE positions 3

- Determines the magnitude of the torque in TWO positions.
AND
- Determines the direction of rotation in TWO positions 2

- Provides some relevant working 1

Question 32

The average distance between two nucleons is approximately 1×10^{-15} m. Thus, as can be seen in Graph 1 and Graph 2, the magnitude of the gravitational force between two nucleons is approximately -2.2×10^{-34} N, and the magnitude of the electrostatic force between two protons is approximately 230 N. The magnitude of the electrostatic force, which is repulsive, is greater than the magnitude of the gravitational force, which is attractive, by a factor of approximately 10^{36} . Given that the magnitude of the electrostatic force is significantly greater than that of the gravitational force, it would be expected that protons would repel each other, thus rendering the nucleus unstable. However, due to the strong nuclear force, atoms are stable. The strong nuclear force exists between nucleons and involves the exchange of π -mesons. At the average distance between nucleons, the strong nuclear force is a strong, attractive force that overcomes the repulsion caused by the electrostatic force, thus accounting for the stability of the nucleus. The addition of neutrons creates a greater strong nuclear force with no increase in the electrostatic force; hence, a higher number of neutrons is required for larger nuclei to remain stable.

Mod 8 From the Universe to the Atom (PH12–5, 12–6, 12–15)

Bands 4–6

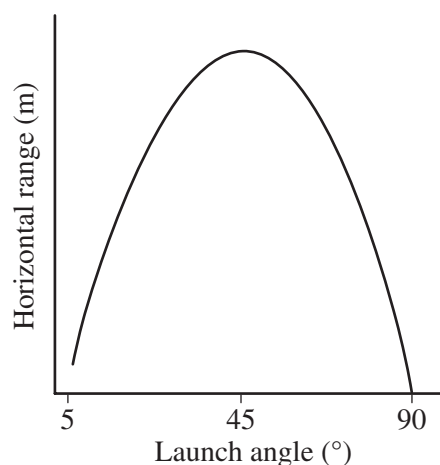
- Refers to data from BOTH graphs.
AND
 - Compares the gravitational AND electrostatic force between nucleons.
AND
 - Identifies that the difference between the magnitudes of the gravitational force and electrostatic force should make the atom unstable.
AND
 - Describes the strong nuclear force.
AND
 - Explains how the strong nuclear force overcomes electrostatic repulsion5
-
- Refers to data from ONE graph.
AND
 - Compares the gravitational AND electrostatic force between nucleons.
AND
 - Describes the strong nuclear force.
AND
 - Explains how the strong nuclear force overcomes electrostatic repulsion4
-
- Compares the gravitational AND electrostatic force between nucleons.
AND
 - Describes the strong nuclear force.
AND
 - Explains how the strong nuclear force overcomes electrostatic repulsion3
-
- Describes the gravitational AND electrostatic force between nucleons.
AND
 - Describes the strong nuclear force.2
-
- Provides some relevant information1

Question 33*For example:*

Projectile motion can be modelled using simulations. By analysing the primary data from a simulation, the relationship between the launch angle (independent variable) and the horizontal range (dependent variable) can be investigated.

A simulation can be used to conduct a number of trials where the launch angle of a projectile is changed and the resultant horizontal ranges are recorded. For example, launch angles could be increased by increments of 5° , in the range of 5° – 90° .

The data can then be plotted on a graph, as shown. The graph shows that as the launch angle increases from 5° to 45° , the horizontal range increases; then, as the launch angle increases from 45° to 90° , the horizontal range decreases. Thus, the relationship is parabolic.



Note: Accept other appropriate models.

Mod 5 Advanced Mechanics (PH12–4, 12–5, 12–7, 12–12)

Bands 4–5

- Identifies ONE appropriate method of modelling projectile motion.
AND
- Outlines the suggested model.
AND
- Identifies the primary data obtained.
AND
- Discusses the relationship between launch angle and horizontal range.
AND
- Sketches an appropriate graph. 5

- Any FOUR of the above points. 4

- Any THREE of the above points. 3

- Any TWO of the above points. 2

- Provides some relevant information 1

Question 34

- (a) In Millikan's experiment, oil droplets were charged using X-rays and then suspended in an electric field produced by two metal electrodes. The field strength was varied to create an upward electric force that balanced the downward gravitational force, allowing the charged oil droplets to become suspended. This showed that the charged oil droplets were interacting with the electric field. The magnitude of the electric field, the magnitude of gravity and the masses of the oil droplets were known. Millikan used these values to calculate the charges of the oil droplets, according to the following formula.

$$F_E = F_g$$

$$qE = mg$$

$$q = \frac{mg}{E}$$

In Millikan's formula, q was consistently calculated as an integer multiple of approximately 1.602×10^{-19} C. Hence, Millikan concluded that the lowest common denominator of the recorded charges was equal to the minimum possible charge, and thus the charge of an electron.

Mod 8 From the Universe to the Atom (PH12-5, 12-6, 12-7, 12-15)

Bands 4-5

- | | |
|--|---|
| <ul style="list-style-type: none"> • Outlines the experiment in detail.
AND • Explains how the experiment determined the charge of the electron | 5 |
| <hr/> | |
| <ul style="list-style-type: none"> • Outlines the experiment.
AND • Explains how the experiment determined the charge of the electron | 4 |
| <hr/> | |
| <ul style="list-style-type: none"> • Outlines the experiment.
AND • States that the experiment determined the charge of the electron | 3 |
| <hr/> | |
| <ul style="list-style-type: none"> • Describes the experiment with limited detail.
AND • States that the experiment determined the charge of an electron | 2 |
| <hr/> | |
| <ul style="list-style-type: none"> • Provides some relevant information | 1 |

- (b) As the positively charged ions enter the uniform magnetic field, a magnetic force acts on the ions. This causes the ions to undergo uniform circular motion, according to the formula $F_c = \frac{mv^2}{r}$.
- The ions deflect in a circular path, accelerating at a rate that is inversely proportional to their masses (centripetal acceleration). As the ions have the same charge and velocity, their radii of curvature will be proportional to their masses. By measuring these radii, the masses of the ions can be determined.

Mod 5 Advanced Mechanics (PH12–7)

Mod 6 Electromagnetism (PH12–4, 12–5)

Bands 5–6

- Explains how the magnetic force results in circular motion.
AND
 - Explains that the centripetal acceleration of the ions is inversely proportional to their masses.
AND
 - Explains how the ions' radii of curvature can be used to determine their masses3
-
- Any TWO of the above points.2
-
- Provides some relevant information1